

Adding, Subtracting, and Multiplying Polynomials

Answer Key

Algebra 1

Quick Answers

1. $4x^2 - 3x + 5$

2. $5x^2 - 10x + 10$

3. $-12x^5 + 6x^3 - 15x^2$

4. $x^2 + 2x - 24$

5. $2x^3 + 5x^2 - 5$

6. $4x^2 - 2x$

7. $6x^2 - 7x - 20$

8. $4x^2y - 7y^2$

9. $8x^2y^2 - 12xy^3$

10. $x^2 - 6x + 9$

11. 36

12. $x^3 - 2x^2 - 4x + 14$

Common wrong answers

11. *If they got -2: multiplied the first terms and the last terms only, skipping the outer and inner products of FOIL*

1. Add $(3x^2 + 5x - 2) + (x^2 - 8x + 7)$.

Addition needs no sign change — drop the parentheses and collect like terms by degree.

x^2 terms: $3x^2 + x^2 = 4x^2$. x terms: $5x - 8x = -3x$. constants: $-2 + 7 = 5$.

$4x^2 - 3x + 5$

2. Subtract $(7x^2 - 4x + 1) - (2x^2 + 6x - 9)$.

The minus sign applies to *every* term of the second polynomial, so rewrite it as an addition first:

$$(7x^2 - 4x + 1) + (-2x^2 - 6x + 9)$$

Now collect: $7x^2 - 2x^2 = 5x^2$, $-4x - 6x = -10x$, $1 + 9 = 10$.

$5x^2 - 10x + 10$

Grading note: an answer of $5x^2 + 2x - 8$ means only the first term of the second polynomial was negated. Insist on writing the rewritten sum line.

3. Multiply $-3x^2(4x^3 - 2x + 5)$.

Distribute the monomial across all three terms. Multiply coefficients, add exponents on x , and carry the negative sign through every product.

$$-3x^2 \cdot 4x^3 = -12x^5, \quad -3x^2 \cdot (-2x) = 6x^3, \quad -3x^2 \cdot 5 = -15x^2.$$

$-12x^5 + 6x^3 - 15x^2$

4. Multiply $(x + 6)(x - 4)$.

Every term of the first factor times every term of the second (FOIL):

$$x \cdot x + x \cdot (-4) + 6 \cdot x + 6 \cdot (-4) = x^2 - 4x + 6x - 24$$

The two middle terms are like terms: $-4x + 6x = 2x$.

$$\boxed{x^2 + 2x - 24}$$

5. Add $(2x^3 - x + 4) + (5x^2 + x - 9)$.

Neither polynomial has all four degrees, so line the terms up by degree and leave a gap where a degree is missing.

x^3 : only $2x^3$. x^2 : only $5x^2$. x : $-x + x = 0$, so the x term vanishes. constants: $4 - 9 = -5$.

$$\boxed{2x^3 + 5x^2 - 5}$$

6. Subtract $(4x^3 + x^2 - 6) - (4x^3 - 3x^2 + 2x - 6)$.

Distribute the minus sign:

$$4x^3 + x^2 - 6 - 4x^3 + 3x^2 - 2x + 6$$

The x^3 terms cancel because they are identical and one is being subtracted from the other; the constants cancel for the same reason, since $-6 - (-6) = 0$.

What is left: $x^2 + 3x^2 = 4x^2$ and the lone $-2x$.

$$\boxed{4x^2 - 2x}$$

7. Multiply $(2x - 5)(3x + 4)$.

$$2x \cdot 3x + 2x \cdot 4 + (-5) \cdot 3x + (-5) \cdot 4 = 6x^2 + 8x - 15x - 20$$

Combine the middle terms: $8x - 15x = -7x$.

$$\boxed{6x^2 - 7x - 20}$$

8. Add $(3x^2y - 5xy + 2y^2) + (x^2y + 5xy - 9y^2)$.

With two variables, like terms must match on *both* letters and both exponents. x^2y and xy are not like terms.

$$3x^2y + x^2y = 4x^2y, \quad -5xy + 5xy = 0, \quad 2y^2 - 9y^2 = -7y^2.$$

$$\boxed{4x^2y - 7y^2}$$

9. Multiply $4xy^2(2x - 3y)$.

Distribute, multiplying coefficients and adding exponents on each variable separately.

$$4xy^2 \cdot 2x = 8x^2y^2 \quad (\text{the } x \text{ exponents add: } 1 + 1 = 2).$$

$$4xy^2 \cdot (-3y) = -12xy^3 \quad (\text{the } y \text{ exponents add: } 2 + 1 = 3).$$

$$\boxed{8x^2y^2 - 12xy^3}$$

10. Why is $(x - 3)^2 \neq x^2 - 9$?

Squaring a binomial means multiplying it by itself, and the distributive property produces *four* products, not two. Squaring each term separately throws away the two middle products.

$$(x - 3)^2 = (x - 3)(x - 3) = x^2 - 3x - 3x + 9$$

$$\boxed{x^2 - 6x + 9}$$

A one-line numeric disproof is worth showing: at $x = 5$ the correct expression gives $(5 - 3)^2 = 4$, while $x^2 - 9$ gives 16.

Grading note: $x^2 - 9$ is the expansion of $(x - 3)(x + 3)$, a different product. Students who confuse the two are pattern-matching on the numbers instead of distributing.

11. Multiply $(3x - 2)(x + 7)$ and evaluate at $x = 2$.

$$3x \cdot x + 3x \cdot 7 + (-2) \cdot x + (-2) \cdot 7 = 3x^2 + 21x - 2x - 14 = 3x^2 + 19x - 14$$

Substitute $x = 2$: $3(4) + 19(2) - 14 = 12 + 38 - 14$.

$$\boxed{36}$$

Grading note: the classmate's -2 comes from multiplying only the first terms and only the last terms — $3x \cdot x - 2 \cdot 7$, which is $3x^2 - 14$, and at $x = 2$ that is $12 - 14 = -2$. The outer and inner products were skipped, which is exactly the $19x$ that went missing.

12. Synthesis: simplify $(x + 2)(x^2 - 3x + 5) - (x - 1)(x + 4)$.

First product — distribute x and then 2 across the trinomial:

$$(x + 2)(x^2 - 3x + 5) = x^3 - 3x^2 + 5x + 2x^2 - 6x + 10 = x^3 - x^2 - x + 10$$

Second product:

$$(x - 1)(x + 4) = x^2 + 4x - x - 4 = x^2 + 3x - 4$$

Now subtract the second from the first, distributing the minus sign across all three of its terms:

$$x^3 - x^2 - x + 10 - x^2 - 3x + 4$$

Collect by degree: x^3 stays; $-x^2 - x^2 = -2x^2$; $-x - 3x = -4x$; $10 + 4 = 14$.

$$\boxed{x^3 - 2x^2 - 4x + 14}$$

Grading note: the most common slip here is subtracting only the x^2 term of the second product. Check the constant: $+14$, not $+6$.